INCH-POUND
MIL-DTL-21337A
3 MARCH 2023
SUPERSEDING
MIL-N-21337
10 APRIL 1968

DETAIL SPECIFICATION

NUTS, PLAIN, ROUND, RETAINING, BALL AND ROLLER BEARINGS

Reactivated after 3 March 2023 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers procurement requirement for slotted retaining round plain nuts for use in assemblies with ball and roller bearings.

1.2 Classification. The nuts shall be of the following types and classes as specified:

Types:

Type I — Regular series

Class 1 - Carbon steel (plain)
Class 2 - Carbon steel (zinc electroplated)
Class 3 - (NOT for new design) Carbon steel (cadmium[1/] plated)
Class 4 - Corrosion resisting steel (Passivated)
Class 5 - Carbon steel (zinc-nickel plated)

[1/] Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 (Class 5 above) is a suitable replacement with design authority approval.

Type II — Light series

Class 1 - Alloy steel (Phosphate coated)

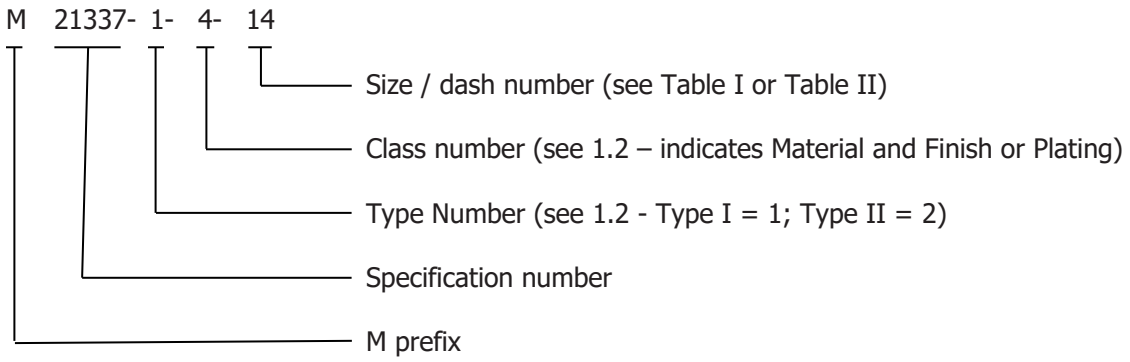
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AMSC N/A FSC 5310

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1.3 Part or Identifying Number (PIN). PINs to be used for nuts, plain, round, retaining, ball and roller bearings acquired to this specification are created as follows:



Examples of PIN:

M21337-1-4-08	Nut, Type I (regular series), Class 4 (CRES, passivated with no post treatment), 2.250 outside diameter.
M21337-1-5-14	Nut, Type I (regular series), Class 5 (carbon steel, Zinc-Nickel plate, Type K16), 3.625 outside diameter.

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-16232	-	Phosphate Coating, Heavy, Manganese or Zinc Base
MIL-PRF-32647	-	Zinc-Nickel Electroplating for Fasteners

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	-	Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B18.18	-	Quality Assurance for Fasteners
ASME B46.1	-	Surface Texture, (Surface Roughness, Waviness and Lay)

(Copies of these documents are available online at https://www.asme.org/.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A959	-	Standard Guide for Specifying Harmonized Standard Grade Compositions for Wrought Stainless Steels
ASTM A967/A967M	-	Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts
ASTM E1282	-	Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials
ASTM E1444/E1444M	-	Standard Practice for Magnetic Particle Testing for Aerospace
ASTM F1941/F1941M	-	Standard Specification for Electrodeposited Coatings on Mechanical Fasteners, Inch and Metric

(Copies of these documents are available online at https://www.astm.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331	-	Zero Acceptance Number Sampling Plans - Fifth Edition

(Copies of these documents are available online at https://www.asq.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AIR4127	-	Steel: Chemical Composition and Hardenability
SAE AMS2700	-	Passivation of Corrosion Resistant Steels
SAE AMS6322	-	Steel Bars, Forgings, and Rings 0.50Cr – 0.55Ni – 0.25Mo (0.38 – 0.43C) (SAE 8740) (Composition similar to UNS G87400)
SAE AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)
SAE J403	-	Chemical Compositions of SAE Carbon Steels
SAE J404	-	Chemical Compositions of SAE Alloy Steels

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.2 Materials. Materials used in the manufacture of nuts shall be homogeneous in structure, free from cracks, pipes, seams, laminations, inclusion of nonmetallic impurities and such other defects as would render the material unsuitable for its intended purpose.

3.2.1 Carbon steel. Type I, classes 1, 2, 3, and 5 nuts shall be made from carbon steel conforming to SAE J403 steel numbers SAE 1005 to SAE 1035 (UNS G10050 to UNS G10350) inclusive.

3.2.2 Corrosion-resisting steel. Type I, class 4 nuts shall be fabricated from corrosion-resisting steel conforming to ASTM A959, steel numbers 302 (UNS S30200), 303 (UNS S30300), or 304 (UNS S30400).

3.2.3 Alloy steel. Type II-Class 1 nuts shall be made from alloy steel conforming to SAE AMS6322 steel number SAE 8740 (UNS G87400) or alloy steel conforming to SAE J404 steel number SAE 8742 (UNS G87420).

3.3 Threads. Threads shall be Unified Inch Screw Threads (UNS) in accordance with ASME B1.1.

3.4 Finish and tolerances.

3.4.1 Protective finish, plating, or coating and treatment. Unless otherwise specified in the contract or order, nuts shall be finished, plated, coated, or treated in accordance with 1.2.

3.4.1.1 Zinc Plating. Type I class 2 nuts shall be zinc electroplated in accordance with ASTM F1941/F1941M Fe/Zn-Ni 5BNS.

3.4.1.2 Cadmium Plating. Type I class 3 nuts shall be cadmium[1/] plated in accordance with SAE AMS-QQ-P-416, Type II, Class 3.

[1/] Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 (Class 5) is a suitable replacement with design authority approval.

3.4.1.3 Passivation. Type I class 4 nuts shall be cleaned, descaled and passivated in accordance with SAE AMS2700 Method 1, Type 6, 7, or 8, or Method 2; or ASTM A967/A967M, Nitric treatment 2-5, or Citric treatment 1-5. Passivation treatments containing hexavalent chromium compounds shall be prohibited.

3.4.1.4 Zinc-Nickel Plating. Type I class 5 nuts shall be zinc-nickel plated in accordance with MIL-PRF-32647, Type K16.

3.4.1.5. Phosphate coating. Type II class 1 nuts shall be phosphate coated in accordance with MIL-DTL-16232, Type Z, Class 3.

3.4.2 Surface roughness. Surface roughness of types I and II nuts shall not be greater than a roughness height rating of 125 except as specified in figure 2. Surface roughness shall be interpreted in accordance with the provisions of ASME B46.1.

3.4.3 Tolerances. Tolerances after finishes, plating, coating, or treatment is applied for Type I and Type II nuts shall be plus or minus 0.010-inch for linear dimensions and plus or minus 2 degrees for angular dimensions except as specified on Figure 1 and Table I or Figure 2 and Table II, as applicable.

3.5 Sharp edges. Unless otherwise specified herein, sharp edges shall be rounded, chamfered, or broken not less than 0.003 inch and not more than 0.015 inch.

3.6 Designs, sizes, and dimensions. Unless otherwise specified in the contract or order, nuts shall conform to the designs, sizes, and dimensions specified on Figure 1 and Table I or Figure 2 and Table II, as applicable. Type I nuts shall have either 4 or 8 slots, at the option of the manufacturer. Type II nuts shall have 8 slots.

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3.7 Examination of Class I nut characteristics. Class I nuts sampled in accordance with 4.3.2 shall be examined in accordance with Figure 1 and Table I for the following:

a. Thickness
b. Outside Diameter
c. Face Diameter
d. Face parallelism
e. Chamfer of Threads
f. Slot Dimensions
g. Thread Data
h. Major Diameter
i. Width Across Flats

3.8 Examination of Class II nut characteristics. Class II nuts sampled in accordance with 4.3.2 shall be examined in accordance with Figure 2 and Table II for the following:

a. Thickness
b. Outside Diameter
c. Face Diameter
d. Face parallelism
e. C'Sink Diameter
f. Slot Dimensions
g. Thread Data
h. Major Diameter
i. Width Across Flats

3.9 Hardness.

3.9.1 Carbon steel. The hardness of carbon steel nuts shall be Rockwell B66 minimum to Rockwell B92 maximum.

3.9.2 Corrosion-resisting steel. Corrosion-resisting steel nuts shall be in the annealed condition and at a hardness of Rockwell B90 maximum.

3.9.3 Alloy steel. Alloy steel nuts shall be hardened by quenching in oil from the hardening temperature and tempered to a Rockwell "C" hardness of 26 to 32.

3.10 Magnetic particle testing. Alloy and carbon steel nuts shall pass the magnetic particle inspection of 4.4.2.

3.11 Parallelism of faces. The runout and parallelism of sides of the nut when tested in accordance with the procedures of 4.5.2.1 shall conform to the tolerances specified in figures 1 or 2 as applicable.

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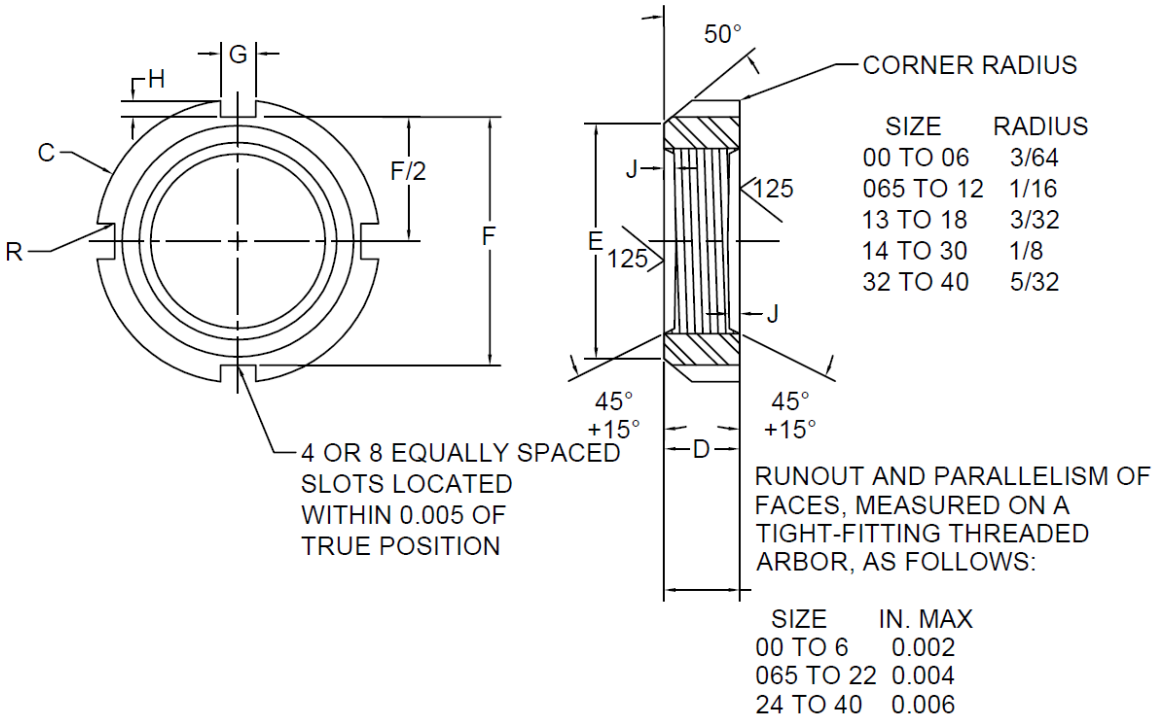


Figure 1. Type I, Nuts

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TABLE I. Type I, Nuts

SIZE / DASH NUM.	D		C	E		J	SLOT DIMENSIONS					THREAD DATA					MAJOR DIA.
							F	G		H	R						
	THICKNESS		OUTSIDE DIA.	FACE DIA.			DIA.	WIDTH				THREADS PER INCH (UNS)	MINOR DIA.		PITCH DIA.		
	MIN	MAX	+0.005 -0.015	MIN	MAX	CHAMP OF THREAD	+0.010 -0.020	MIN	MAX	NOM	MAX		MIN	MAX	MIN	MAX	MIN
00	0.209	0.229	0.750	0.605	0.625	0.031	0.625	0.120	0.130	0.062	0.010	32	0.3572	0.3606	0.3707	0.3733	0.391
01	0.303	0.323	0.875	0.699	0.719	0.031	0.750	0.120	0.130	0.062	0.010	32	0.4352	0.4386	0.4487	0.4513	0.469
02	0.303	0.323	1.000	0.793	0.813	0.031	0.812	0.120	0.130	0.094	0.010	32	0.5522	0.5556	0.5657	0.5687	0.586
03	0.334	0.354	1.125	0.918	0.938	0.031	0.937	0.120	0.130	0.094	0.010	32	0.6302	0.6336	0.6437	0.6467	0.664
04	0.365	0.385	1.375	1.105	1.125	0.031	1.187	0.178	0.198	0.094	0.015	32	0.7472	0.7506	0.7607	0.7641	0.781
05	0.396	0.416	1.562	1.261	1.281	0.031	1.375	0.178	0.198	0.094	0.015	32	0.9352	0.9386	0.9487	0.9521	0.969
06	0.396	0.416	1.750	1.480	1.500	0.047	1.562	0.178	0.198	0.094	0.015	18	1.1129	1.1189	1.1369	1.1409	1.173
065	0.428	0.448	2.062	1.793	1.813	0.047	1.875	0.178	0.198	0.094	0.015	18	1.2524	1.2584	1.2764	1.2804	1.3125
07	0.428	0.448	2.062	1.793	1.813	0.047	1.875	0.178	0.198	0.094	0.015	18	1.3159	1.3219	1.3399	1.3439	1.376
08	0.428	0.448	2.250	1.980	2.000	0.047	2.062	0.240	0.260	0.094	0.020	18	1.5029	1.5089	1.5269	1.5314	1.563
09	0.428	0.448	2.531	2.261	2.281	0.047	2.343	0.240	0.260	0.094	0.020	18	1.7069	1.7129	1.7309	1.7354	1.767
10	0.490	0.510	2.687	2.418	2.438	0.047	2.500	0.240	0.260	0.094	0.020	18	1.9069	1.9129	1.9309	1.9354	1.967
11	0.490	0.510	2.968	2.636	2.656	0.047	2.718	0.240	0.260	0.125	0.020	18	2.0969	2.1029	2.1209	2.1260	2.157
12	0.521	0.541	3.156	2.824	2.844	0.047	2.906	0.240	0.260	0.125	0.020	18	2.2999	2.3059	2.3239	2.3290	2.360
13	0.553	0.573	3.375	3.043	3.063	0.047	3.125	0.240	0.260	0.125	0.020	18	2.4879	2.4939	2.5119	2.5170	2.548
14	0.553	0.573	3.625	3.283	3.313	0.047	3.375	0.240	0.260	0.125	0.020	18	2.6909	2.6969	2.7149	2.7200	2.751
15	0.584	0.604	3.875	3.533	3.563	0.078	3.625	0.360	0.385	0.125	0.025	12	2.8428	2.8518	2.8789	2.8843	2.933
155	0.584	0.604	3.875	3.533	3.563	0.078	3.625	0.360	0.385	0.125	0.025	11	2.8428	2.8518	2.8789	2.8843	2.933
16	0.584	0.604	4.156	3.814	3.844	0.078	3.906	0.360	0.385	0.125	0.025	12	3.0468	3.0558	3.0829	3.0888	3.137
165	0.584	0.604	4.156	3.814	3.844	0.078	3.906	0.360	0.385	0.125	0.025	11	3.0468	3.0558	3.0829	3.0888	3.137
17	0.615	0.635	4.406	4.001	4.031	0.078	4.093	0.360	0.385	0.156	0.025	12	3.2498	3.2588	3.2859	3.2933	3.340
175	0.615	0.635	4.406	4.001	4.031	0.078	4.093	0.360	0.385	0.156	0.025	11	3.2498	3.2588	3.2859	3.2933	3.340
18	0.678	0.698	4.656	4.251	4.281	0.078	4.343	0.360	0.385	0.156	0.025	12	3.4368	3.4458	3.4729	3.4803	3.527
185	0.678	0.698	4.656	4.251	4.281	0.078	4.343	0.360	0.385	0.156	0.025	11	3.4368	3.4458	3.4729	3.4803	3.527
19	0.709	0.729	4.937	4.533	4.563	0.078	4.625	0.360	0.385	0.156	0.025	12	3.6398	3.6488	3.6759	3.6833	3.730
195	0.709	0.729	4.937	4.533	4.563	0.078	4.625	0.360	0.385	0.156	0.025	11	3.6398	3.6488	3.6759	3.6833	3.730
20	0.735	0.760	5.187	4.783	4.813	0.078	4.875	0.360	0.385	0.156	0.025	12	3.8278	3.8368	3.8639	3.8713	3.918
205	0.735	0.760	5.187	4.783	4.813	0.078	4.875	0.360	0.385	0.156	0.025	11	3.8278	3.8368	3.8639	3.8713	3.918
21	0.735	0.760	5.437	4.970	5.000	0.078	5.062	0.485	0.510	0.188	0.030	12	4.0318	4.0408	4.0679	4.0762	4.122
22	0.766	0.791	5.718	5.251	5.281	0.078	5.343	0.485	0.510	0.188	0.030	12	4.2348	4.2438	4.2709	4.2792	4.325
24	0.798	0.823	6.125	5.658	5.688	0.078	5.750	0.485	0.510	0.188	0.030	12	4.6258	4.6348	4.6619	4.6702	4.716
26	0.860	0.885	6.750	6.158	6.188	0.078	6.250	0.610	0.635	0.250	0.030	12	5.0158	5.0248	5.0519	5.0602	5.106
28	0.923	0.948	7.093	6.501	6.531	0.078	6.593	0.610	0.635	0.250	0.030	12	5.4068	5.4158	5.4429	5.4512	5.497
285	0.923	0.948	7.093	6.501	6.531	0.078	6.593	0.610	0.635	0.250	0.030	11	5.4068	5.4158	5.4429	5.4512	5.497
30	0.954	0.979	7.687	7.033	7.063	0.078	7.125	0.610	0.635	0.281	0.030	12	5.7978	5.8068	5.8339	5.8422	5.888
32	1.016	1.041	8.062	7.398	7.438	0.094	7.500	0.610	0.635	0.281	0.030	8	6.1487	6.1622	6.2028	6.2119	6.284
34	1.048	1.073	8.656	7.991	8.031	0.094	8.093	0.610	0.635	0.281	0.030	8	6.5237	6.5372	6.5778	6.5869	6.659
36	1.079	1.104	9.062	8.335	8.375	0.094	8.427	0.735	0.760	0.312	0.030	8	6.9307	6.9442	6.9848	6.9939	7.066
38	1.110	1.135	9.468	8.741	8.781	0.094	8.843	0.735	0.760	0.312	0.030	8	7.3367	7.3502	7.3908	7.3999	7.472
385	1.110	1.135	9.468	8.741	8.781	0.094	8.843	0.735	0.760	0.312	0.030	11	7.3367	7.3502	7.3908	7.3999	7.472
40	1.173	1.198	9.843	9.116	9.156	0.094	9.218	0.735	0.760	0.312	0.030	8	7.7117	7.7252	7.7658	7.7772	7.847

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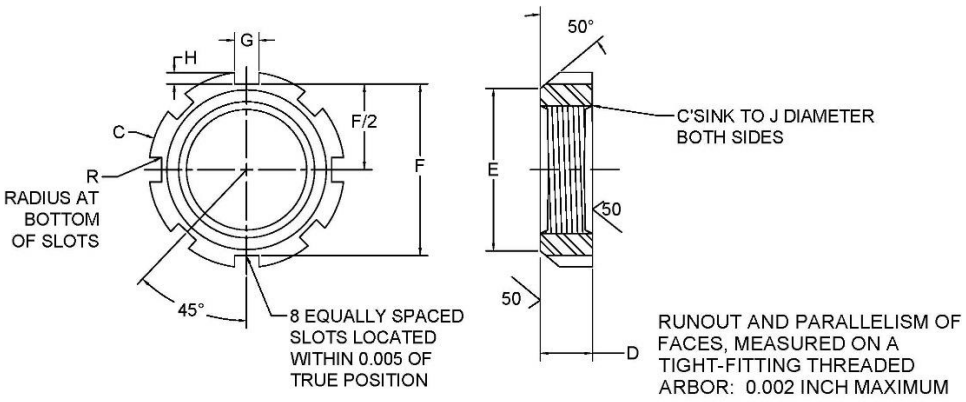


Figure 2. Type II, Nuts

TABLE II. Type II, Nuts

SIZE / DASH	D THICKNESS	C OUTSIDE DIA. +0.005 -0.015	E FACE DIA. +0.000 -0.020	J C'SINK DIA.	SLOT DIMENSIONS F	G	H	R	THREAD DATA THREADS PER INCH	MINOR DIA. MIN	MINOR DIA. MAX	PITCH DIA. MIN	PITCH DIA. MAX	MAJOR DIA. MIN
A00	0.188	0.719	0.531	0.401	0.594	0.125	0.062	0.010	32	0.3572	0.3606	0.3707	0.3733	0.391
A01	0.188	0.812	0.625	0.479	0.688	0.125	0.062	0.010	32	0.4352	0.4386	0.4487	0.4513	0.469
A02	0.219	0.938	0.750	0.596	0.812	0.125	0.062	0.010	32	0.5522	0.5556	0.5657	0.5687	0.586
A03	0.219	1.062	0.875	0.674	0.938	0.125	0.062	0.010	32	0.6302	0.6336	0.6437	0.6467	0.664
A04	0.250	1.156	0.969	0.791	1.031	0.188	0.062	0.015	32	0.7472	0.7506	0.7607	0.7641	0.781
A05	0.250	1.375	1.188	0.979	1.250	0.188	0.062	0.015	32	0.9352	0.9386	0.9487	0.9521	0.969
A06	0.281	1.688	1.438	1.183	1.500	0.188	0.094	0.015	18	1.1129	1.1189	1.1369	1.1409	1.173
A07	0.281	1.906	1.656	1.386	1.719	0.188	0.094	0.015	18	1.3159	1.3219	1.3399	1.3439	1.376
A08	0.281	2.094	1.844	1.573	1.906	0.250	0.094	0.020	18	1.5029	1.5089	1.5269	1.5314	1.563
A09	0.312	2.312	2.062	1.777	2.125	0.250	0.094	0.020	18	1.7069	1.7129	1.7309	1.7354	1.767
A10	0.312	2.500	2.250	1.977	2.312	0.250	0.094	0.020	18	1.9069	1.9129	1.9309	1.9354	1.967
A11	0.312	2.750	2.500	2.167	2.562	0.250	0.094	0.020	18	2.0969	2.1029	2.1209	2.1260	2.157
A12	0.344	2.938	2.688	2.370	2.750	0.250	0.094	0.020	18	2.2999	2.3059	2.3239	2.3290	2.360
A13	0.344	3.125	2.875	2.558	2.937	0.250	0.094	0.020	18	2.4879	2.4939	2.5119	2.5170	2.548
A14	0.344	3.344	3.094	2.761	3.156	0.250	0.094	0.020	18	2.6909	2.6969	2.7149	2.7200	2.751
A15	0.375	3.688	3.375	2.943	3.438	0.375	0.125	0.025	12	2.8428	2.8518	2.8789	2.8843	2.933
A16	0.375	3.906	3.594	3.147	3.656	0.375	0.125	0.025	12	3.0468	3.0558	3.0829	3.0888	3.137
A17	0.375	4.125	3.812	3.350	3.875	0.375	0.125	0.025	12	3.2498	3.2588	3.2859	3.2933	3.340
A18	0.375	4.312	4.000	3.537	4.062	0.375	0.125	0.025	12	3.4368	3.4458	3.4729	3.4803	3.527
A19	0.406	4.531	4.219	3.740	4.281	0.375	0.125	0.025	12	3.6398	3.6488	3.6759	3.6833	3.730
A20	0.406	4.719	4.406	3.928	4.469	0.375	0.125	0.025	12	3.8278	3.8368	3.8639	3.8713	3.918
A21	0.406	4.906	4.594	4.132	4.656	0.500	0.125	0.030	12	4.0318	4.0408	4.0679	4.0762	4.122
A22	0.406	5.125	4.812	4.335	4.875	0.500	0.125	0.030	12	4.2348	4.2438	4.2709	4.2792	4.325

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3.12 Identification marking. Each nut shall be permanently and legibly marked with the manufacturer's name or trademark and part number by a steel stamp, acid or electric etching, or engraving in accordance with the provisions of MIL-STD-130.

3.13 Workmanship. Workmanship shall be in accordance with high grade commercial practice governing this type of material which will ensure compliance with all the requirements of this specification. Each nut shall be free from hanging burrs and slivers, gouges, porosity, objectionable scale or any other defects which may adversely affect the nut's serviceability.

4 VERIFICATION

4.1 General. When this specification and the applicable documents thereto specifies action by the Government inspector, it shall be performed by or under the supervision of the Government inspector.

4.2 Lot. A lot shall consist of all nuts of the same type, class and size submitted for acceptance inspection and test at one time.

4.3 Sampling.

4.3.1 Sampling for lot acceptance inspection. Unless otherwise specified in thew contract (see 6.2), for sampling for lot acceptance inspection, an "accept at zero" (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. For purposes of inspection, as specifies in section 4.4.1, samples shall be selected from each lot in accordance with MIL-STD-1916 or ASQ H1331 using the verification levels or index values specified in table III.

4.3.2 For lot acceptance tests.

4.3.2.1Sampling for composition and hardness. Unless otherwise specified in the contract (see 6.2), for sampling for composition and hardness tests, an "accept at zero" (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. For purposes of composition and hardness testing, as specifies in 4.5.1.3 and 4.5.1.4, samples shall be selected from each lot in accordance with MIL-STD-1916 verification level I or ASQ H1331 index value 10.

4.4 Inspection.

4.4.1 Lot acceptance inspection. Sampling nuts selected in accordance with 4.3.1 shall be visually and dimensionally inspected in accordance with Table III for conformance to the requirements of this specification and the contract or order.

4.4.2 Magnetic particle inspection. The presence of cracks, seams, inclusions, and flaws in alloy and carbon steel nuts selected in accordance with 4.3.1, shall be determined by magnetic particle inspection unless visual inspection discloses defects which would preclude the necessity for magnetic particle inspection. Magnetic particle inspection shall be performed in accordance with ASTM E1444/E1444M.

4.4.3 Place of inspection. Unless otherwise specified in the contract or order, inspection and test shall be conducted at the place of manufacture under ambient conditions.

4.5 Tests.

4.5.1 Lot acceptance tests.

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4.5.1.1 Side runout and parallelism of faces. Each sample nut selected in accordance with 4.3.1 shall be subjected to a runout and parallelism test according with the procedure of 4.5.2.1 to determine conformance to the tolerance specified in figures 1 and 2 as applicable.

4.5.1.2 Examination of nut characteristics. Each sample nut shall be subjected to examination of characteristics (see 3.7 and 3.8) in accordance with the tolerance specified in Figures 1 or 2, as applicable to Class, to determine conformance. Failure to meet the tolerances shall be cause for rejection of the lot.

4.5.1.3 Chemical composition. Sample nuts from each material, lot, and class shall be selected and have a chemical composition analysis performed in accordance with ASTM E1282. Failure to meet the chemical composition required for the specific material (see 3.2), shall be cause for rejection of the lot.

4.5.1.4 Hardness. Sample nuts selected shall be tested for hardness in accordance with procedures of SAE AIR4127. Failure to meet the hardness criteria shall be cause for rejection of the lot.

4.5.2 Test and inspection procedures.

4.5.2.1 Side runout and parallelism of faces. The sample nut shall be mounted on a tight-fitting threaded arbor, (a tapered arbor is not acceptable for this purpose), and a calibrated indicator "applied against the contact or chamfered side of the nut (see fig. 3). The thread dimensions of the arbor employed shall conform to the thread data and tolerances specified in tables I and II for the applicable type and class of nut.

4.6 Rejected lot. A rejected lot maybe resubmitted for Government acceptance in accordance with the provisions of Standard MIL-STD-1916.

Table III. Classification of defects

Categorization and defects	Method of inspection
Major – Verification Level IV, Index value 1.5	
101. Runout and parallelism of faces	See 4.5.2.1
102. Thread pitch and minor diameters	Gage
103. "1?" diameter (see Tables I and II)	*SMI or gage
104. Cracks, seams, inclusions, and flaws	Visual and magnetic particle inspection
Minor – Verification Level II, Index value 4.0	
201. "D" thickness (see Tables I and II)	*SMI or gage
202. Surface Roughness	Visual
203. "C" outside diameter	*SMI or gage
204. Protective coating	Visual
205. Slots equally spaced within 0.006 of basic position S	Gage
206. Burrs and sharp edges	Visual
*Standard measuring instrument e.g., scale, calipers, micrometers, etc.	

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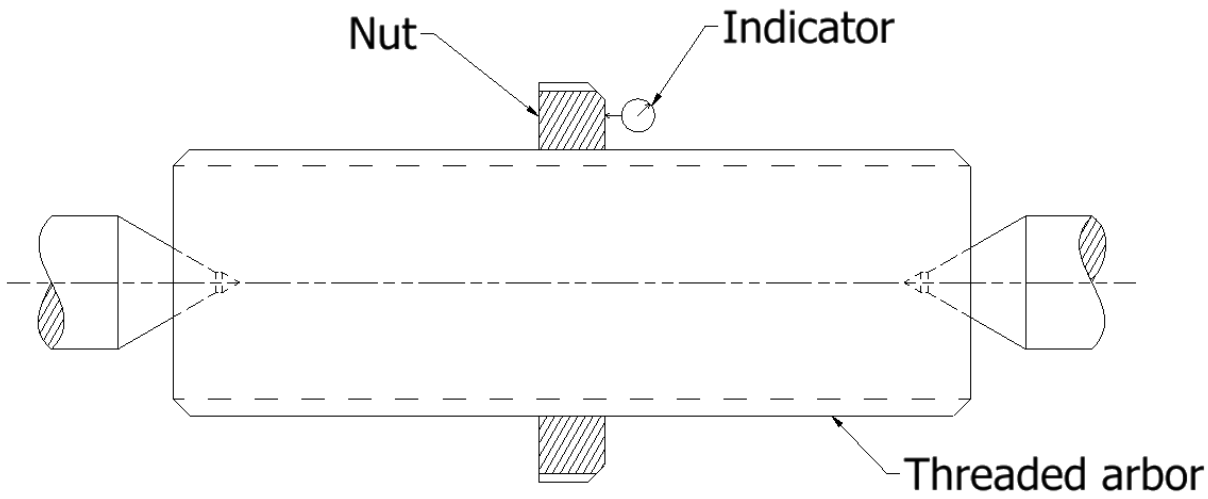


Figure 3. Mounting of Nut on Arbor

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Slotted retaining nuts, covered by this specification are primarily intended for use with pronged lockwashers in securing and locking ball and roller bearings in position against shaft shoulders. This specification covers the requirements for MS19067 and MS19068.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size of nut required.
c. Protective finish required.
d. Special marking (if required).
e. Sampling method for inspection, composition, and hardness tests (4.3.1 & 4.3.2.1).
f. Packaging requirements (5.1)
g. PIN (1.3)

6.3 Definitions. Terms designated standard names, and associated terminology for examination and testing shall be interpreted in accordance with definitions outlined in MIL-STD-1916.

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6.4 Subject term (key word) listing.

a. Plating
b. Finish
c. Zinc
d. Zinc-Nickel
e. Passivate
f. Carbon Steel
g. Alloy Steel
h. Corrosion Resistant Steel
i. CRES
j. Cadmium

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – OS
Air Force – 99
DLA – GS

Preparing activity:
DLA – GS7

(Project 5310-2023-007)

Review activities:

Army – AR, MI
Navy – MC, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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